

Budget Tracking System

For: Addis Ababa City Administration (AACCA)
Bureau of Finance

Requirement Analysis Document (Business Process Study)

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Version 1.0

March 19, 2026

Revision History

Date	Version	Description	Author
March 19, 2026	1.0	Draft for documenting and guide the development team on partial view of the business process.	Dr. Dagmawi Lemma dagmaw.Lemma@gmail.com

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1. Introduction

1.1. Purpose

The purpose of this document is to define the functional and non-functional requirements for the automation of budget tracking. The system aims to ensure 100% traceability of funds from the Central Bureau of Finance down to specific object codes in sector-specific programs.

1.2. Scope

This system covers the lifecycle from **Budget Code Generation** to **Expenditure Validation** and **Real-time Budget Status Reporting**.

2. Process Overview (BPMN)

The Business Process is sought into two parts:

- The Account Setting and Opening Balance done by the AACA Budget Office and AACA Budget Allocation Wing (see Figure 2.1)
- The Budget Consumption and Tracking (see Figure 2.2)

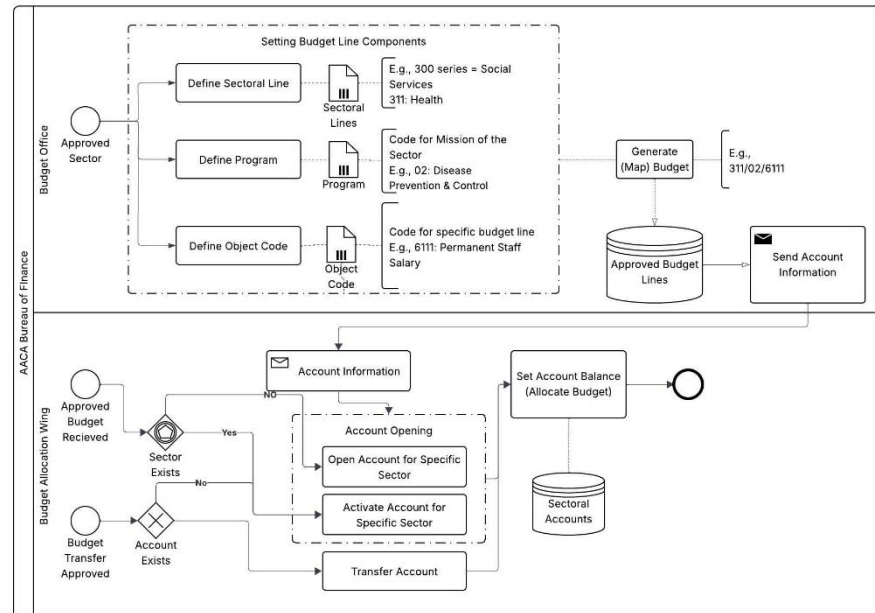


Figure 2-1: BPMN for Account Setting and Opening Balance

3. Functional Requirements

3.1. Budget Line Initialization

Req ID	Requirement Description	Priority
FR-01	The system shall generate a unique 4-segment Traceability ID for every budget line (Jurisdiction/Sector/Program/Object).	High
FR-02	The system must initialize ledger balances based on the Approved Budget Bill provided by the City Council.	High

3.2. Automated Validation Logic

Req ID	Business Rule	System Action
BR-01	Balance Check	Reject any transaction where: Requested Amount > Available Balance
BR-02	Code Integrity	Validate that the Object Code (e.g., 6111) is compatible with the Sector Code (e.g., 311) Validate that the Program Code (e.g., 02) is applicable/defined for the respective Sector

4. User Roles & Permissions

To ensure security and accountability, the following roles are defined:

Role	Permissions	Responsibilities
System Admin	Full Access	Manage Chart of Accounts (COA) and System Configuration.
Central Budget Office (CBO)	Write/Edit	Allocate funds, generate codes, and approve reallocations.
Sector User (e.g. Health/AAWSA)	Create/View	Initiate spend requests and view sector-specific balances.
Auditor	Read-Only	Access all traceability reports and transaction logs for inspection.

5. Data Dictionary (Standardized Codes)

5.1. Jurisdiction

The jurisdiction refers to the specific level of government or the legal geographic boundary that has the authority to collect revenue and spend a budget, which is the Addis Ababa City Administration and is identified by the first two digits, according to the scope of this system it is 10*.

5.2. Sector Code

The second a 3-digit numeric value represents the sector under the respective jurisdiction, in this case under AACAA. Under AACAA, there are three major sectors represented as follows:

- o **General Government Services (100 Series):** These codes cover the administrative and legislative backbone of the city.
- o **Economic Services (200 Series):** This is where the majority of "Capital Budget" lines are tracked for infrastructure.
- o **Social Services (300 Series):** These sectors focus on human capital and "pro-poor" spending.

5.3. Program

The Core Operational Program, which is mission of the respective Sector is designated by the third element of the budget code, which is a 2-digit number. For example, code 02 indicates the Disease Prevention & Control, which is one of the missions of the Health Bureau (311).

5.4. Object

- **(e.g., 6111):** Basic Salary.

Since it is required to trace **which sector consumes how much of a specific budget line**, this structure is the key element of the budget code structure.

E.g.,

- **Salaries**, are represented by 6111 across all Sectors.
- **Supplies**, are represented by 6231 across all Sectors.

5.5. The Complete Structure

The complete structure of the budget line thus composes all the four codes as shown in the scenario below.

Scenario: *The Addis Ababa City Cabinet decides to expand maternal health services in the Gullele Sub-City. To do this, they need to hire 50 new mid-level health extension workers immediately.*

The Budget Line Involved: 10/311/02/6111 (Translation: Addis Ababa City (10) / Health Bureau (311) / Disease Prevention & Control Program (02) / Basic Salary (6111).)

Example Workflow - Budget Amendment (CBO Level): Suppose the existing budget for 10/311/02/6111 was 500 million ETB. To employ the 50 new staff, the Central Budget Office (CBO)

* It is important to make this code configurable.

must increase this specific line by 15 million ETB (this is assuming the basic salary would be 25000 and the new employees will be engaged for full budget year).